Exercises of Functional Analysis

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1. Why $\sum_{k=1}^{n}(a_k \cdot x^{m_k}) = 0 \implies a_k = 0 \quad \forall \quad 1 \leq k \leq n$?

2. If $T : \mathbb{R} \to \mathbb{R}$ is a function such that $T(x+y) = T(x) + T(y) \quad \forall \quad x, y \in \mathbb{R}$. Assume that T is continuous at $x_0 \in \mathbb{R}$. Prove that T is linear map.

3. Let V, W be two vector spaces and $x(\neq 0) \in V, y \in W$. Then prove that there exists a linear transformation $T : V \to W$ such that $T(x) = y$.

4. Let V, W be two vector spaces over $\mathbb{F}$, then $L(V, W)$ is the set of all linear maps from V to W and addition and scalar multiplication are defined as follows:

$$(T+S)(x) = T(x) + S(x) \quad \forall x \in V, T, S \in L(V, W)$$
$$(\alpha \cdot T)(x) = \alpha \cdot T(x) \quad \forall x \in V, T \in L(V, W), \alpha \in \mathbb{F}$$

   Let $\{e_1, e_2, \ldots, e_n\}$ be a basis of V. Define $f_i : V \to \mathbb{F}$ as $f_i(e_j) = \delta_{ij}$ where δ_{ij} is Kronecker delta. $\{f_1, f_2, \ldots, f_n\}$ is a basis of $V' = L(V, \mathbb{F})$ the algebraic dual of V. Let $x \in V$ and $f(x) = 0 \quad \forall \quad f \in V'$. Prove that $x = 0$.
   Proof : if $x \neq 0$, construct a functional $g \in V'$ such that $g(x) = 1$.

5. Let $(V, <.,.>)$ be an inner product space. let $\mathcal{A} \subset V$. We say that $\mathcal{A}$ is orthonormal set on V if and only if

   - $< x, y >= 0 \quad \forall \quad x, y \in \mathcal{A}, x \neq y$
   - $< x, x >= 1 \quad \forall \quad x \in \mathcal{A}$

   Prove that $\mathcal{A}$ is linearly independent set on V.

6. Let (X, d) be a metric space and (x_n) be a cauchy sequence in X and (x_{n_k}) be a convergent subsequence of (x_n) such that $\lim_{k\to\infty} x_{n_k} = x$. Prove that $\lim_{n\to\infty} x_n = x$.

7. Prove that every discrete metric space is complete.

8. Let f, g be two functions in $\mathcal{R}[a, b]$(i. e. Riemann Integrable function on [a, b]). we say that $f \sim g$ if and only if

$$\int_a^b |f(x) - g(x)|dx = 0$$

   Prove that $\sim$ is an equivalence relation on $\mathcal{R}[a, b]$.

9. Define

$$l^p(\mathbb{N}) = \{(x_n) \in S(\mathbb{N}); \sum_{n=1}^{\infty} |x_n|^p < \infty\}$$

   for $1 \leq p < \infty$.
   Let p, q be two numbers such that $p, q \in [1, \infty]$ and $p \leq q$.then prove the following:

   - $||(x_n)||_q \leq ||(x_n)||_p \quad \forall \quad (x_n) \in l^p(\mathbb{N})$
   - $l^p(\mathbb{N}) \subset l^q(\mathbb{N})$

10. Prove the following inclusion for $1 \leq p < \infty$:

$$C_{00}(\mathbb{N}) \subset l^p(\mathbb{N}) \subset C_0(\mathbb{N}) \subset C(\mathbb{N}) \subset l^\infty(\mathbb{N})$$

    Also, find $(x_n) \in C_0$ but $(x_n) \notin l^p$ for $1 \leq p < \infty$.

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11. Notation: $e_i = (0, 0, \ldots, 1, \ldots)$ where 1 is at i^{th} position.
clearly, $\{e_i\} \in C_{00}(\mathbb{N}) \quad \forall i \in \mathbb{N}$. Prove that $\{e_i : i \in \mathbb{N}\}$ is a linearly independent set. Hence $C_{00}(\mathbb{N}), l^p(\mathbb{N}), C_0(\mathbb{N}), C(\mathbb{N})$ are infinite dimensional.

12. Let $(X, ||.||)$ be a normed linear space and $A \subseteq X$. Define

$$\begin{aligned} \eta_A(x) &= d(x, A) \\ &= \inf_{y \in A} ||x - y|| \end{aligned}$$

Then prove that $\eta_A : X \to \mathbb{R}$ is a uniformly continuous function.

13. Let $(X, ||.||)$ be normed linear space and $Y \subseteq X$ and $x \in X \backslash Y$. Then prove the following:
    (a) $x \in \overline{Y} \iff d(x, Y) = 0$
    (b) Let Y be a closed subspace of X and $x \in X \backslash Y$, Then $d(x, Y) > 0$

14. Let $(X, +, \cdot)$ be a vector space over $\mathbb{F}$. Let

$$\mathcal{N} = \{||.|| : X \to \mathbb{R} \text{ such that } ||.|| \text{ is a norm on } X\}$$

Let $||.||_a, ||.||_b \in \mathcal{N}$. we say that $||.||_a \sim ||.||_b$ if and only if there exists $c, C > 0$ such that $c||x||_a \leq ||x||_b \leq C||x||_a \quad \forall \quad x \in X$.
Prove that $\sim$ is an equivalence relation on $\mathcal{N}$.

15. Prove that following lemma: Let $(X, ||.||_a)$ and $(X, ||.||_b)$ be two normed linear space. Assume that $||.||_a \sim ||.||_b$. Then we have following:
    (a) $lim_{n\to\infty} x_n = x$ in $(X, ||.||_a)$ if and only if $lim_{n\to\infty} x_n = x$ in $(X, ||.||_b)$.
    (b) (x_n) is cauchy in $(X, ||.||_a)$ if and only if (x_n) is cauchy in $(X, ||.||_b)$.
    (c) let $K \subseteq X$. Then K is compact in $(X, ||.||_a)$ if and only if K is compact in $(X, ||.||_b)$. Morever, The map $f : (X, ||.||_a) \to (X, ||.||_b)$ defined as $f(x) = x$ is homeomorphism.

16. Prove that $(\mathbb{F}^n, ||.||_p)$ is a banach space for $1 \leq p \leq \infty$.

17. Prove that $(l^\infty, ||.||_\infty)$ is a banach space.

18. Let K be compact Hausdorff space. Let $C(K) = \{f : K \to \mathbb{F} \mid \text{f is continuous}\}$. We know that $(C(K), +, \cdot)$ is a vector space. Define $||.||_\infty$ as

$$||f||_\infty = \sup_{x \in K} |f(x)|$$

$(C(K), ||.||_\infty)$ is a normed linear space.
Let $f_n \in C(K) \quad \forall \quad n \in \mathbb{N}$ be a cauchy sequence in $(C(K), ||.||_\infty)$.
Define $f : K \to \mathbb{F}$ as

$$f(x) = lim_{n\to\infty} f_n(x) \forall x \in K$$

Prove that $f \in C(K)$ and $||f - f_n||_\infty \to 0$ as $n \to \infty$.

19. Let $(X, ||.||)$ be a normed linear space. Show that $B(x, y)$ and $\overline{B}(x, y)$ are convex sets.

20. Let $0 < p < 1$. Show that $(\mathbb{R}^2, ||.||_p)$ is not a normed linear space.

21. Let X be a normed linear space. Let $A \subseteq X$ then A is bounded if and only if $\exists M > 0$ such that $||x|| \leq M$ for all $x \in A$.

22. Show that S_x is closed and bounded in X, where $(X, ||.||)$ is a normed linear space and $S_x = \{y \in X : ||y - x|| = 1\}$.

23. Let (X, d) be a metric space. let $A \subseteq B \subseteq X$. Then prove the following:
    (a) $B^o = \emptyset \iff \overline{B^c} = X$
    (b) if $\overline{A} = X$ then $\overline{B} = X$.
    (c) if $\overline{B} = X$ and $B \subset \overline{A}$ then $\overline{A} = X$.

24. Prove that $(\mathbb{C}^n, ||.||_2)$ is separable.
(**Hint** : $(Q + iQ)^n$ is countable and dense in $\mathbb{C}^n$)

25. Let (X, d_1) and (Y, d_2) be two metric spaces. Let $f : X \to Y$ be a continuous function. Assume that X is separable metric space, then prove that $(f(X), d_2)$ is separable.

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